

Wealth and Markets: Capital Markets and Dividends

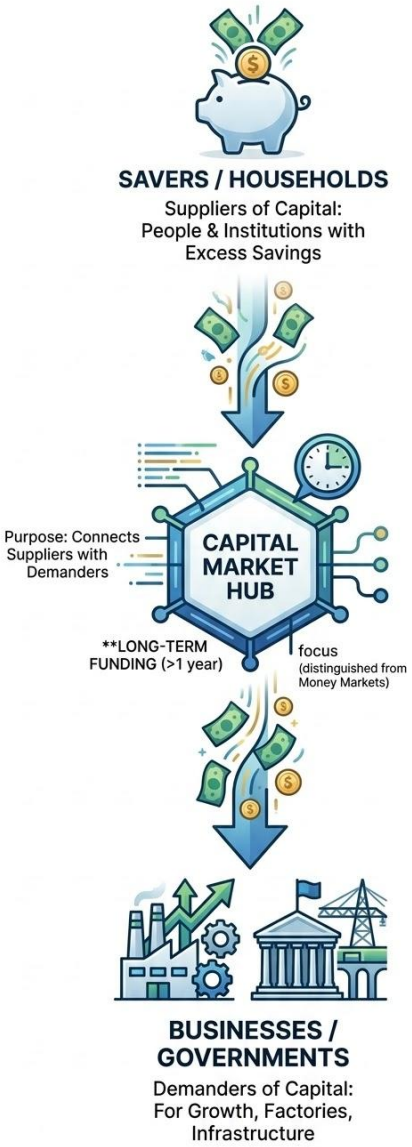
Moving from the Government's Fiscal Policy to the Central Bank's Monetary Policy.

ITGE275 Economics

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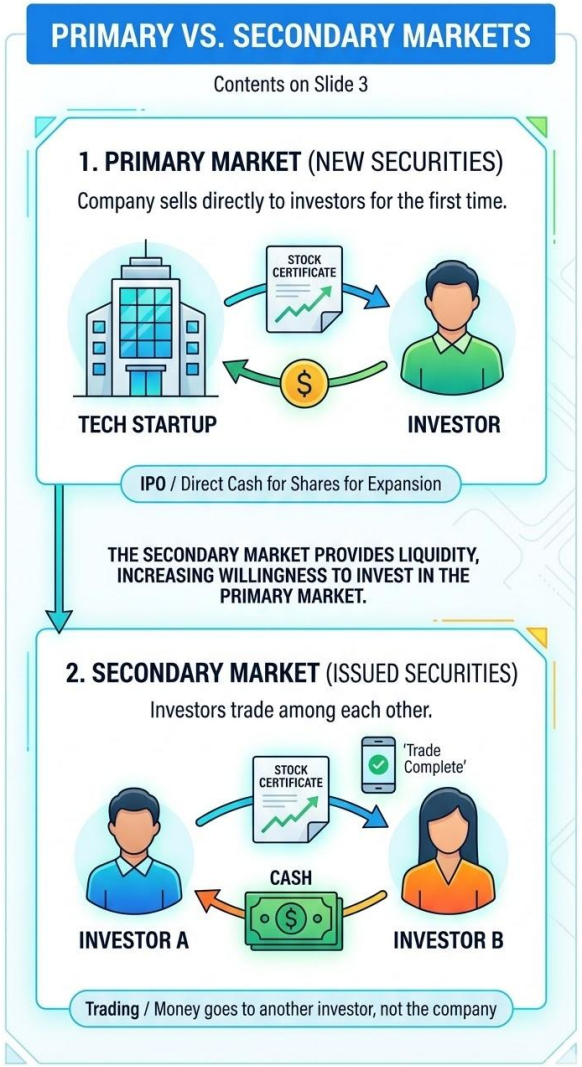
Introduction to the Capital Market

- **Concept:** In previous sessions, we looked at markets where goods and services are traded (like apples or software). The **Capital Market** is where *Money (Capital)* is traded.
- **Purpose:** It connects "Suppliers of Capital" (people or institutions with excess savings) with "Demanders of Capital" (companies or governments that need money to grow, build factories, or fund infrastructure).
- **Time Horizon:** Capital markets specifically deal with *long-term funding* (typically longer than a year), distinguishing them from "Money Markets" which handle short-term borrowing.



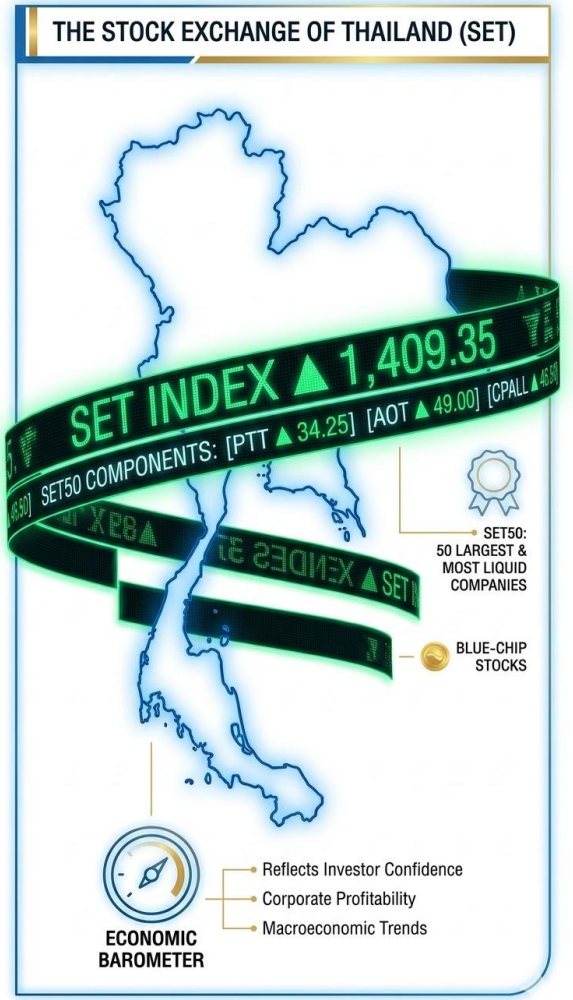
Primary vs. Secondary Markets

- **Primary Market:** Where new securities are created and sold for the very first time.
  - *Example:* An Initial Public Offering (IPO). When a tech startup goes public, it sells shares directly to investors to raise cash for expansion.
- **Secondary Market:** Where previously issued securities are traded among investors.
  - *Example:* Buying Apple stock on your phone today. The money you pay goes to another investor, not to Apple itself.
- **Importance:** The secondary market provides *liquidity*, making investors willing to buy in the primary market because they know they can sell later.



The Stock Exchange of Thailand (SET)

- **Local Context:** The SET Index is a capitalization-weighted index of all common stocks traded on the Stock Exchange of Thailand.
- **Function:** It serves as a barometer for the Thai economy, reflecting investor confidence, corporate profitability, and macroeconomic trends.
- **The SET50:** A sub-index tracking the 50 largest and most liquid companies in Thailand (e.g., PTT, AOT, CPALL). These are often considered "Blue-Chip" stocks.



The Two Pillars of Corporate Finance

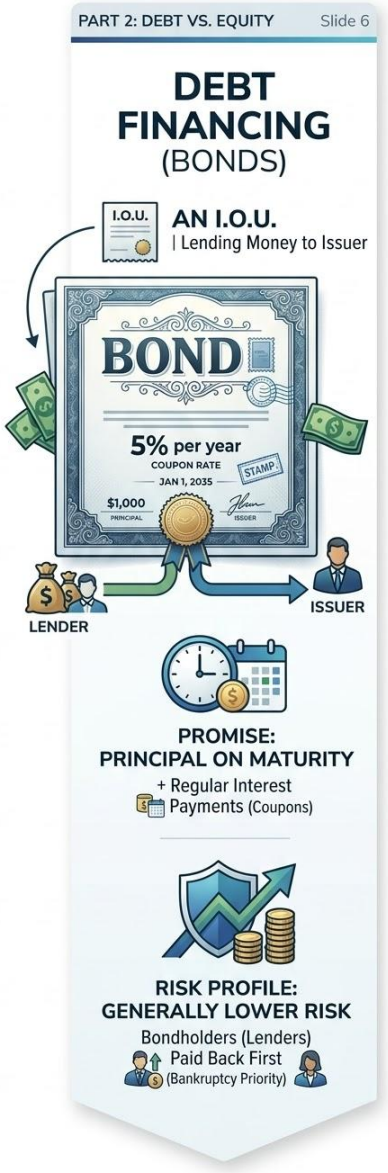
- When a company needs to build a new factory, it cannot just print money. It has two fundamental choices to raise capital:
  - **Debt Financing:** Borrowing money that must be paid back with interest.
  - **Equity Financing:** Selling a piece of ownership in the company.
- Understanding the difference between these two instruments is the foundation of wealth management and corporate strategy.



Part 2: Debt vs. Equity

Debt Financing (Bonds)

- **What is a Bond?** An I.O.U. When you buy a bond, you are lending money to the issuer (corporation or government).
- **The Promise:** The issuer promises to pay you back the principal amount on a specific maturity date, plus regular interest payments (called "coupons") along the way.
- **Risk Profile:** Generally considered lower risk. In the event of a company going bankrupt, bondholders (lenders) are legally required to be paid back *before* stockholders.



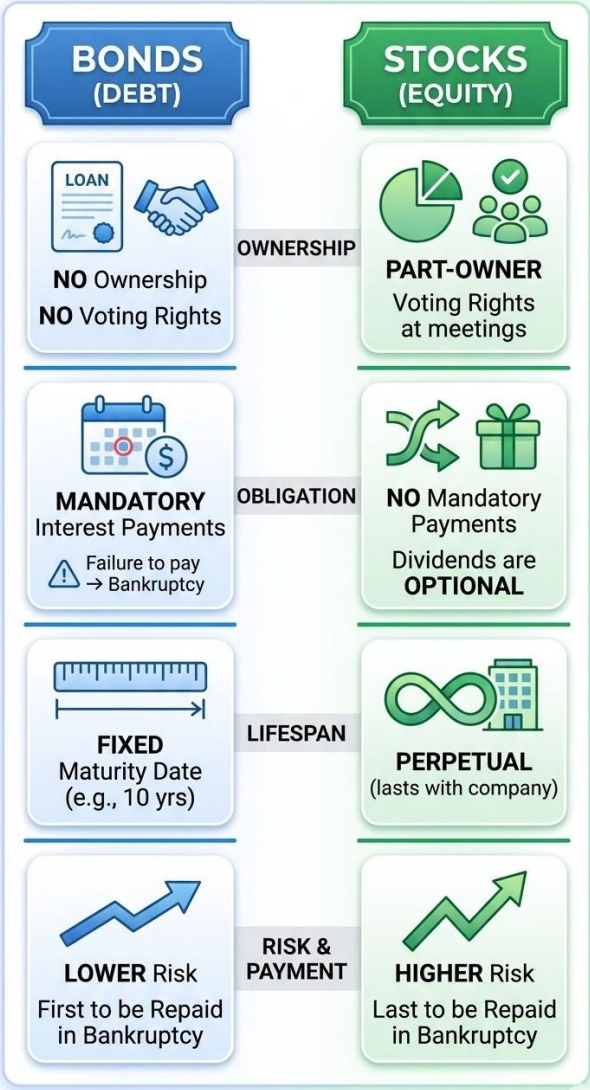
Equity Financing (Stocks/Shares)

- **What is a Stock?** A claim of partial ownership in a corporation.
- **The Reality:** When you buy equity, there is no guarantee you will get your money back, and no guaranteed regular payments.
- **The Upside:** Your return is tied directly to the company's success. If the company becomes the next Amazon, the value of your ownership stake can grow exponentially.
- **Risk Profile:** High risk. Stockholders are at the absolute bottom of the payout ladder if a company goes bankrupt.



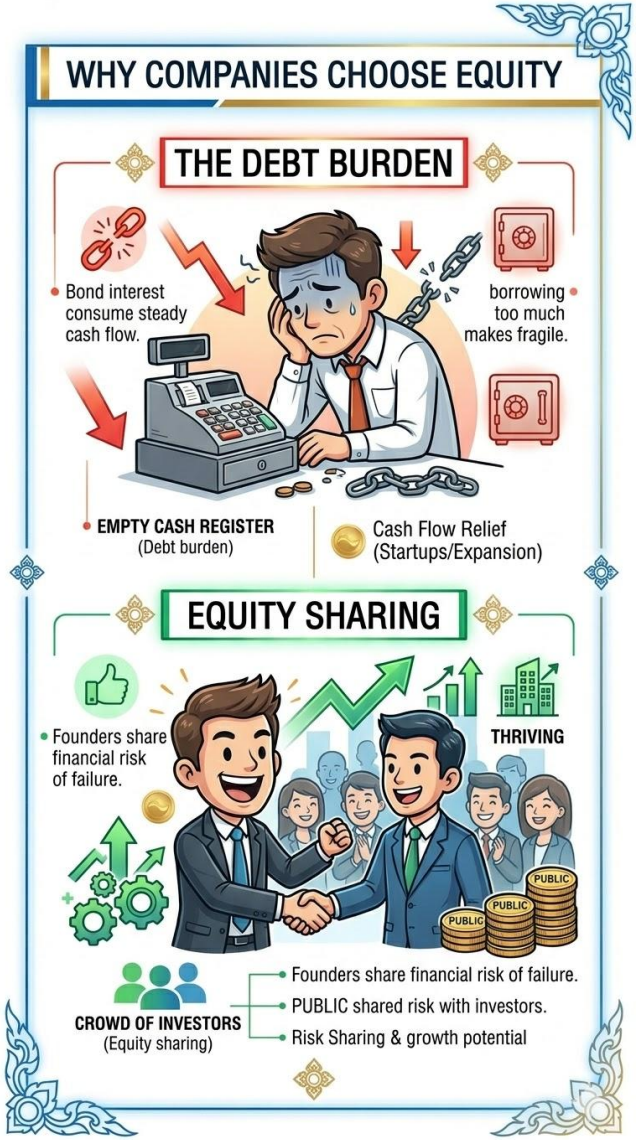
Key Differences: Debt vs. Equity

- **Ownership:** Debt gives you no ownership or voting rights. Equity makes you a part-owner with voting rights at shareholder meetings.
- **Obligation:** Debt requires mandatory interest payments; failure to pay means bankruptcy. Equity has no mandatory payments; dividends are optional.
- **Lifespan:** Debt has a fixed maturity date (e.g., 10 years). Equity lasts as long as the company exists.



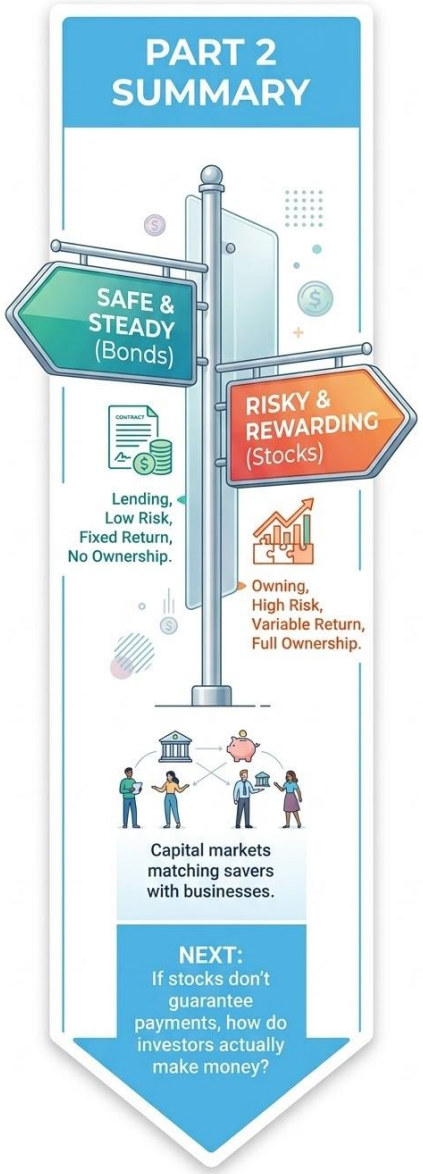
Why Companies Choose Equity

- If equity is more expensive in the long run (sharing profits forever), why do companies issue stock?
- **Cash Flow Relief:** Startups or expanding companies may not have the steady cash flow required to make monthly bond interest payments.
- **Risk Sharing:** By issuing stock, the founders share the financial risk of failure with the public investors.
- **Debt Limits:** Borrowing too much makes a company highly leveraged and fragile during economic downturns.



Part 2 Summary

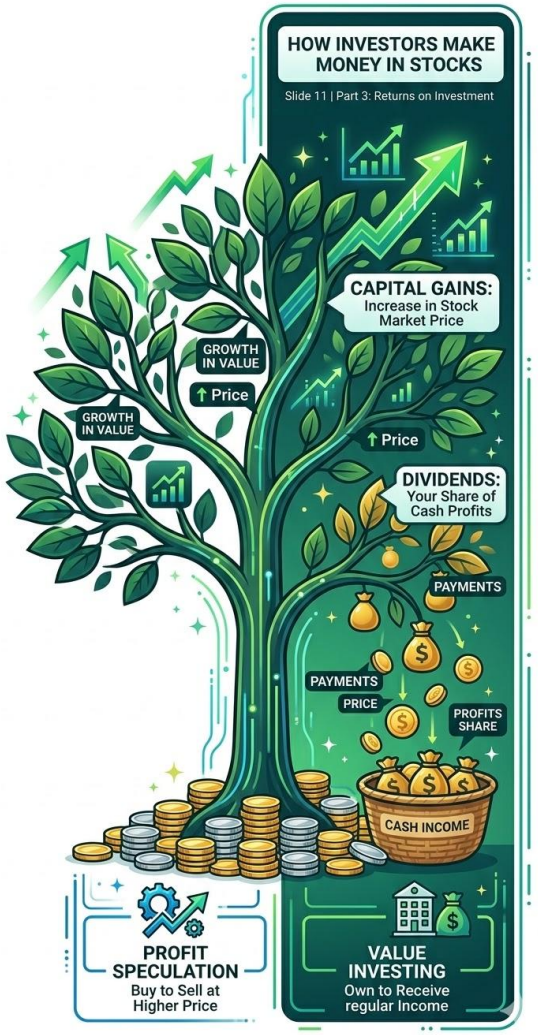
- **Capital Markets** match savers with businesses.
- **Bonds = Lending.** Low risk, fixed return, no ownership.
- **Stocks = Owning.** High risk, variable return, full ownership of profits and losses.
- **Next:** If stocks don't guarantee payments, how do investors actually make money?



Part 3: Returns on Investment

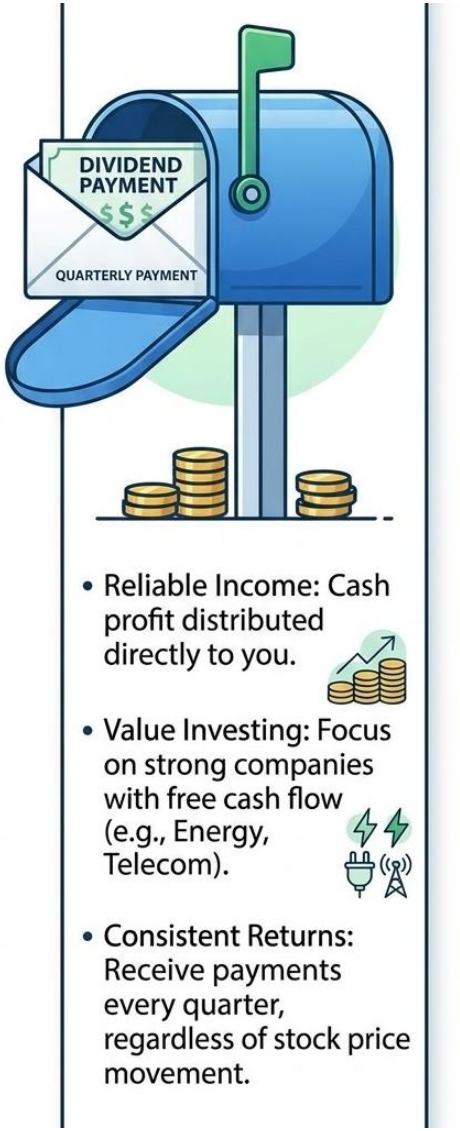
How Investors Make Money in Stocks

- When you own a stock, there are two distinct ways you generate a return on your investment:
  - **Dividends:** Your share of the company's actual, realized cash profits.
  - **Capital Gains:** An increase in the market price of the stock itself.
- These two paths represent two entirely different philosophies of interacting with the financial market: Value Investing vs. Profit Speculation.



Dividends: The Real Profit Sharing

- **Concept:** If a company makes a billion dollars in profit, the Board of Directors can decide to distribute a portion of that cash directly to the shareholders.
- **Value Investing Focus:** Value investors look for established, fundamentally strong companies that generate massive free cash flow and pay reliable dividends (e.g., energy companies, telecom).
- **The Reality:** Even if the stock price goes nowhere for 5 years, a value investor still receives real cash in their bank account every quarter just for holding the asset.



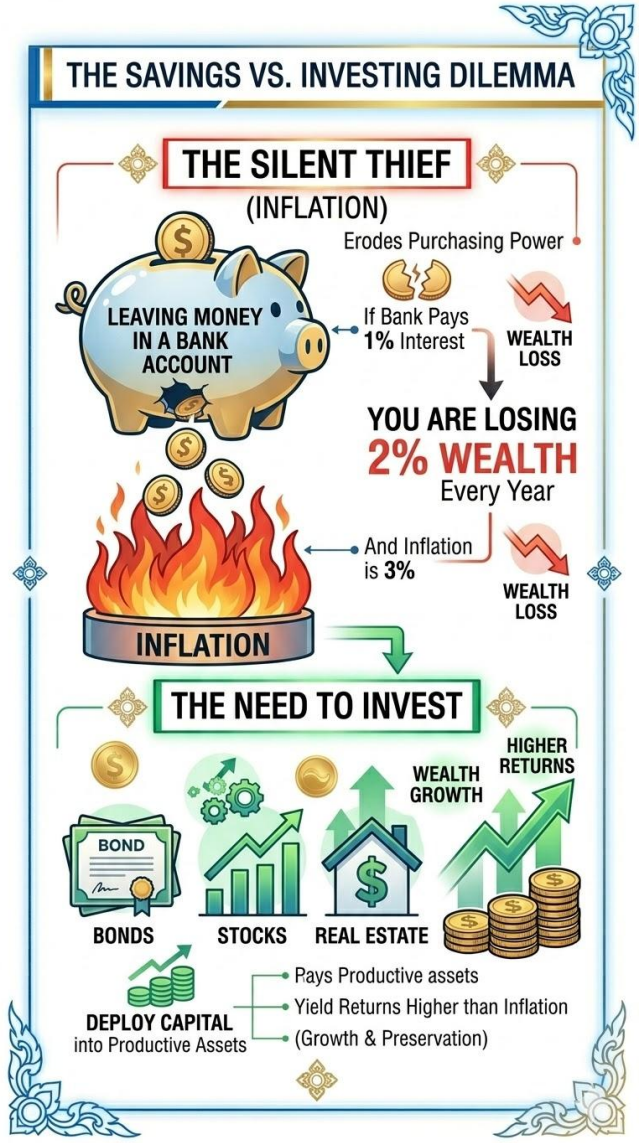
Capital Gains & Profit Speculation

- **Concept:** You buy a stock for 100 THB today, hoping the market will value it at 150 THB next year. The 50 THB difference is your capital gain (if you sell).
- **Speculation Focus:** This relies entirely on the secondary market's perception of the company's future value. It is driven by growth expectations, news, and market psychology.
- **Growth Stocks:** Many tech startups pay *zero* dividends. They reinvest all profits back into the company to grow faster, relying entirely on capital gains to reward investors.



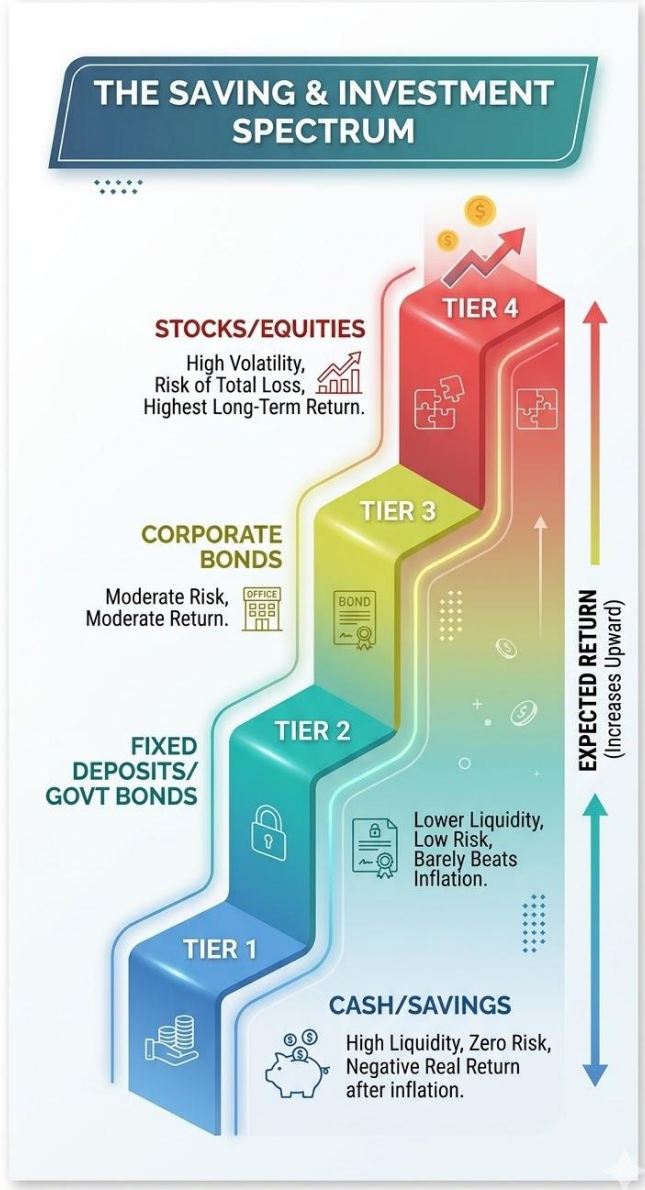
The Savings vs. Investing Dilemma

- Why not just leave money in a bank account?
- **The Silent Thief (Inflation):** As discussed in Session 8, inflation erodes purchasing power. If your bank pays 1% interest and inflation is 3%, you are losing 2% of your wealth every year.
- **The Need to Invest:** To grow, or even just preserve, wealth over the long term, capital must be deployed into productive assets (Bonds, Stocks, Real Estate) that yield returns higher than inflation.



The Spectrum of Saving Options

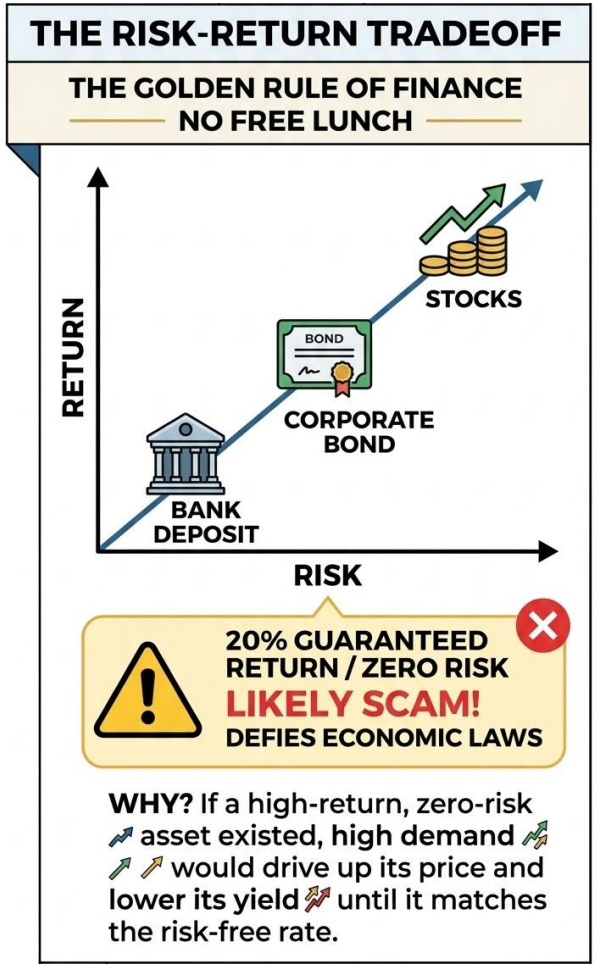
- Investors must choose where to park their capital based on a spectrum of risk.
  - **Tier 1 (Cash/Savings):** High liquidity, virtually zero risk, negative real return after inflation.
  - **Tier 2 (Fixed Deposits/Govt Bonds):** Lower liquidity (locked up for years), low risk, barely beats inflation.
  - **Tier 3 (Corporate Bonds):** Moderate risk (company could default), moderate return.
  - **Tier 4 (Stocks/Equities):** High volatility, risk of total loss, historically the highest long-term return.



Part 4: Risk and the Efficient Market

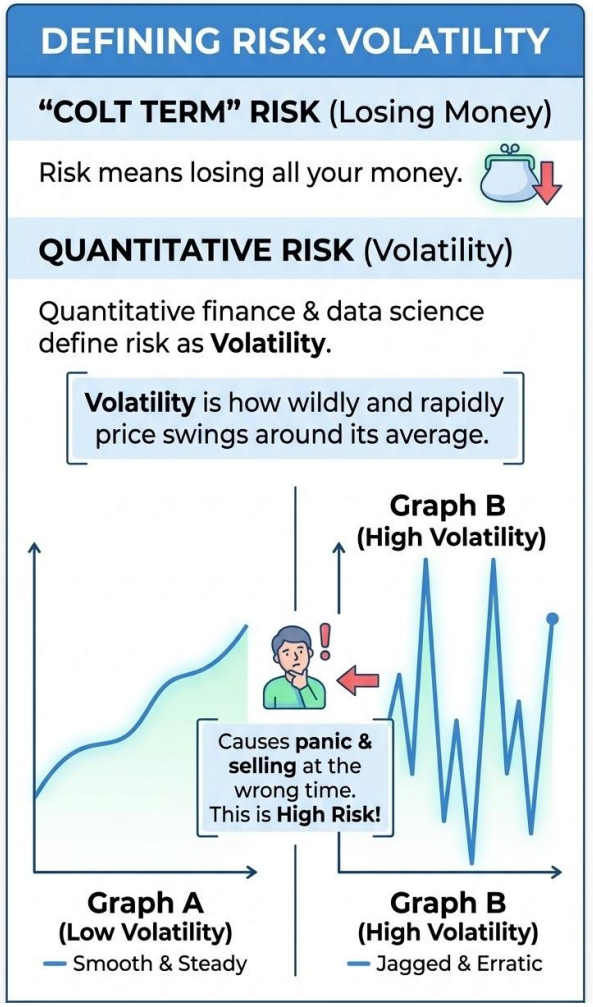
The Risk-Return Tradeoff

- **The Golden Rule of Finance:** There is no such thing as a free lunch. To potentially earn a higher return, you **must** accept a higher level of risk.
- If an investment promises 20% guaranteed returns with zero risk, it defies the fundamental laws of economics (it is likely a scam).
- **Why?** If a high-return, zero-risk asset existed, everyone would buy it, driving its price up and its yield down until it matched the risk-free rate.



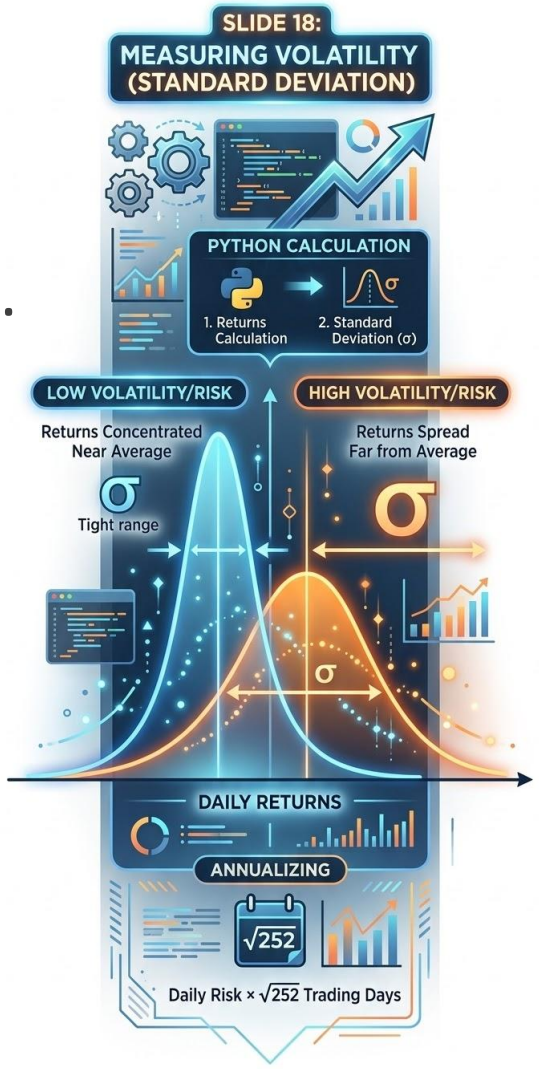
Defining Risk: Volatility

- In colloquial terms, "risk" means losing all your money. In quantitative finance and data science, risk is defined as **Volatility**.
- **Volatility:** How wildly and rapidly the price of an asset swings up and down around its average.
- **The Psychology:** A stock that averages 10% annual returns but frequently drops 30% in a single month is highly volatile. This volatility causes investors to panic and sell at the wrong time.



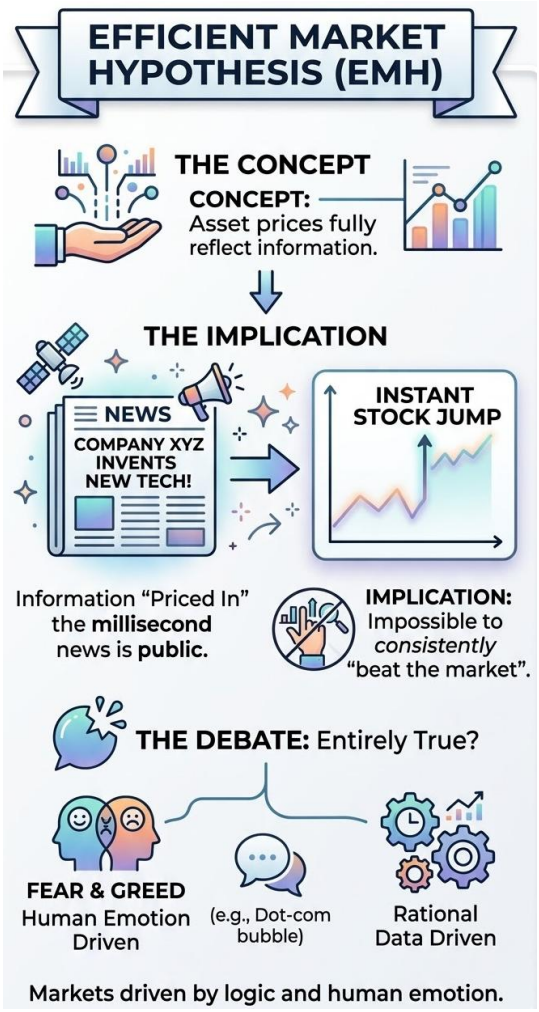
Measuring Volatility (Standard Deviation)

- How do we measure this computationally in our Python notebook?
- **Returns:** We calculate the daily percentage change of a stock.
- **Risk (Standard Deviation):** We calculate the standard deviation (σ) of those daily returns. A higher standard deviation means the daily returns are spread far apart from the average, indicating high risk.
- **Annualizing:** We typically multiply daily risk by the square root of 252 (the number of trading days in a year) to get an Annualized Volatility metric.



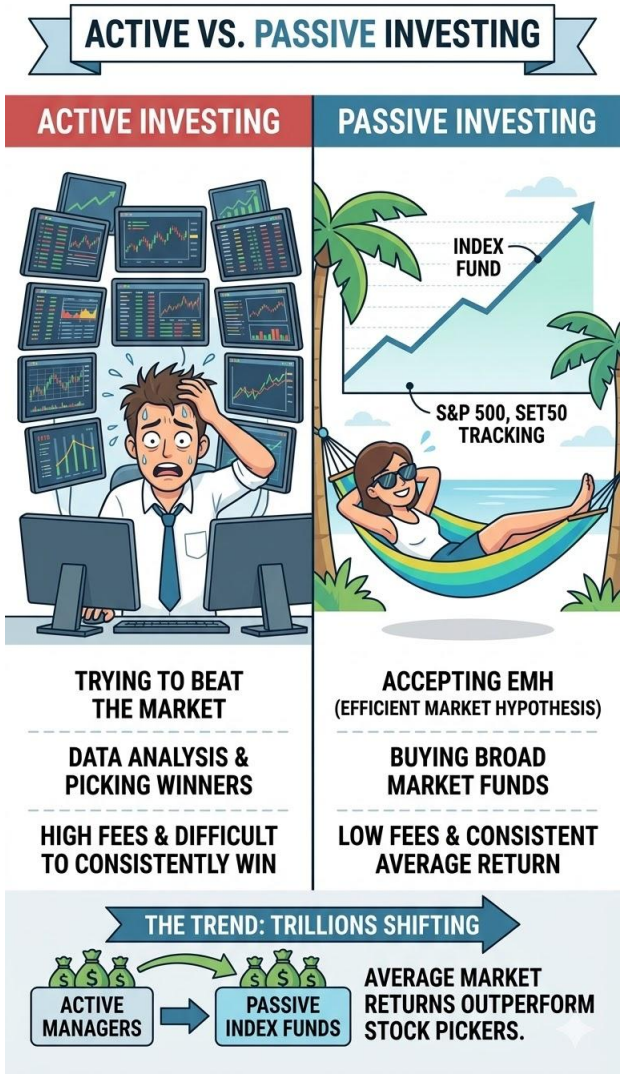
The Efficient Market Hypothesis (EMH)

- **The Concept:** EMH states that asset prices fully reflect all available information.
- **The Implication:** If markets are perfectly efficient, it is impossible to consistently "beat the market" through stock picking or technical analysis, because any good news or bad news is instantly priced into the stock the millisecond it becomes public.
- **The Debate:** Is EMH entirely true? If it were, bubbles (like the dot-com bubble) wouldn't exist. Markets are driven by human emotion (fear and greed) as much as by rational data.



Active vs. Passive Investing

- **Active Investing:** Trying to beat the market by analyzing data and picking winning stocks. (High fees, difficult to consistently win against algorithms).
- **Passive Investing:** Accepting EMH and buying a broad "Index Fund" that tracks the entire market (like the SET50 or S&P 500).
- **The Trend:** Over the last decade, trillions of dollars have moved from active managers to passive index funds because the average market return usually outperforms individual stock pickers.



Part 4 Summary

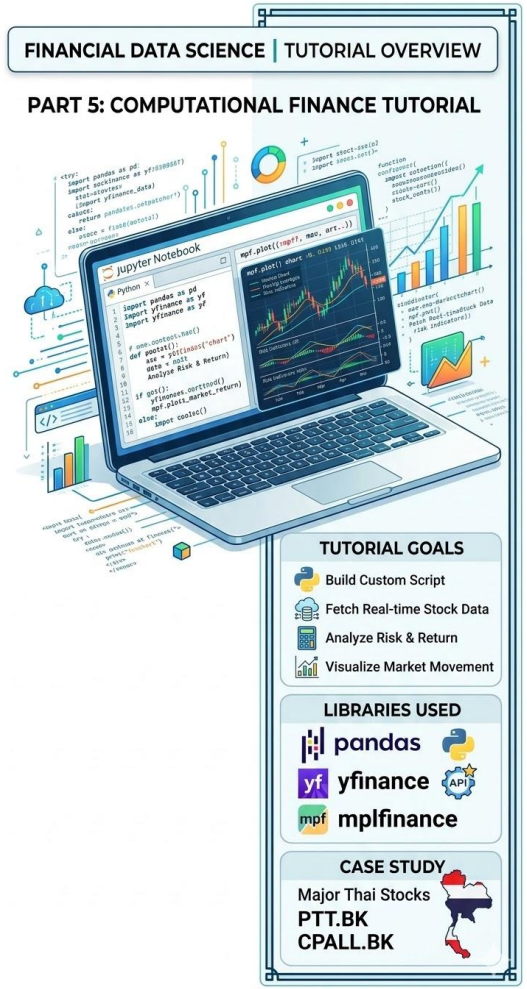
- **Risk = Volatility** (Standard Deviation of returns).
- **Risk-Return Tradeoff:** You cannot get high rewards without enduring high volatility.
- **EMH:** Information is quickly priced in, making speculation difficult.
- **Next:** Let's look at how we calculate and visualize this in Python.



Part 5: Computational Finance Tutorial

Tutorial Overview: Financial Data Science

- **Goal:** Build a script to fetch real stock data, calculate risk/return, and visualize market movements.
- **Tools:**
  - pandas for data manipulation.
  - yfinance to connect to Yahoo Finance's live API.
  - mplfinance for professional financial charting.
- **Case Study:** Analyzing major Thai stocks (PTT.BK, CPALL.BK) to see theory in action.



Fetching Data with yfinance

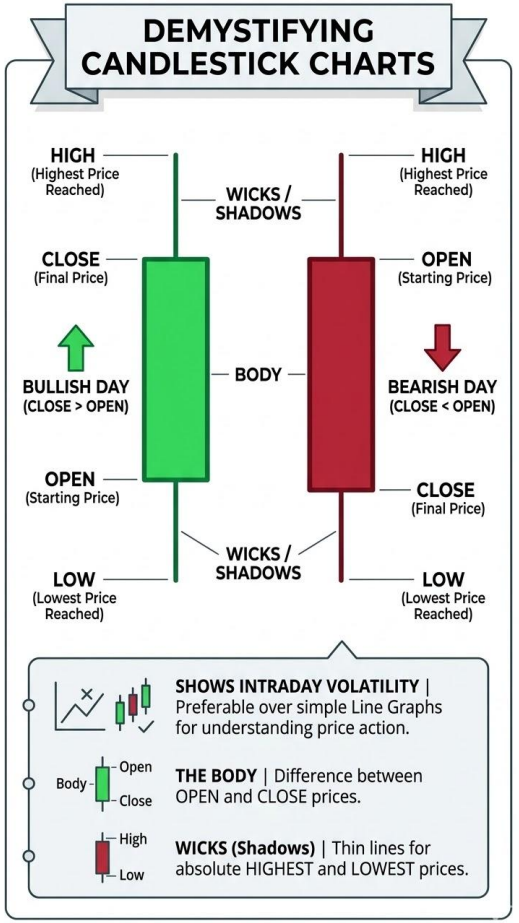
- We no longer rely on static CSV files. We can pull live data from the internet.
- **Code Logic:**
  - ticker = yf.Ticker('PTT.BK')
  - df = ticker.history(start='2020-01-01', end='2024-01-01')
- **Output:** A clean Pandas DataFrame containing the Open, High, Low, Close, and Volume for every single trading day.

Date	Open	High	Low	Close	Volume	Dividends	Stock Splits
2020-01-02 00:00:00+07:00	31.023294	31.899658	30.848021	31.899658	59084700	0.0	0.0
2020-01-03 00:00:00+07:00	32.425481	32.951300	32.250208	32.600754	147370600	0.0	0.0
2020-01-06 00:00:00+07:00	33.126578	33.477124	32.600760	32.951305	91026800	0.0	0.0
2020-01-07 00:00:00+07:00	33.126575	33.126575	32.425484	33.126575	60059600	0.0	0.0
2020-01-08 00:00:00+07:00	33.301842	33.477115	32.776024	32.776024	57759900	0.0	0.0



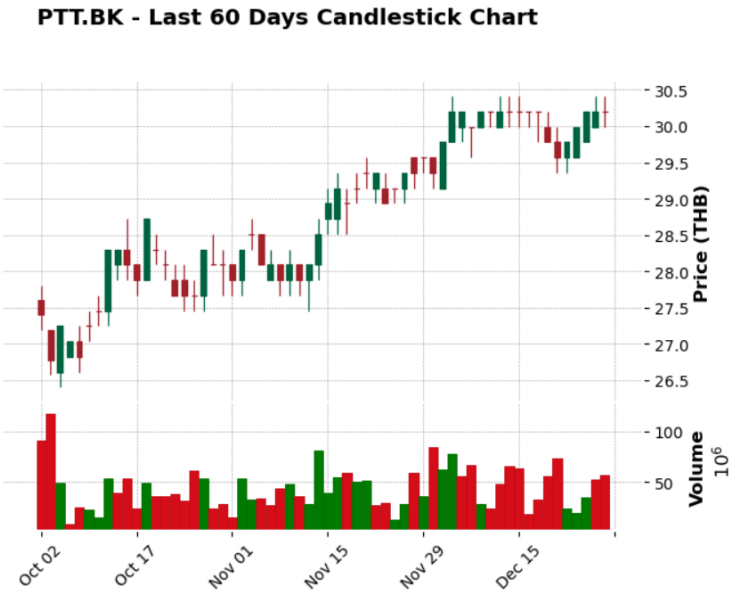
Demystifying Candlestick Charts

- Professional traders use Candlestick charts, not simple line graphs, because they show intraday volatility.
- **The Body:** The thick part represents the difference between the Opening price and the Closing price.
  - *Green/White:* Close > Open (Bullish day).
  - *Red/Black:* Close < Open (Bearish day).
- **The Wicks (Shadows):** The thin lines sticking out top and bottom represent the absolute Highest and Lowest prices reached during that specific day.



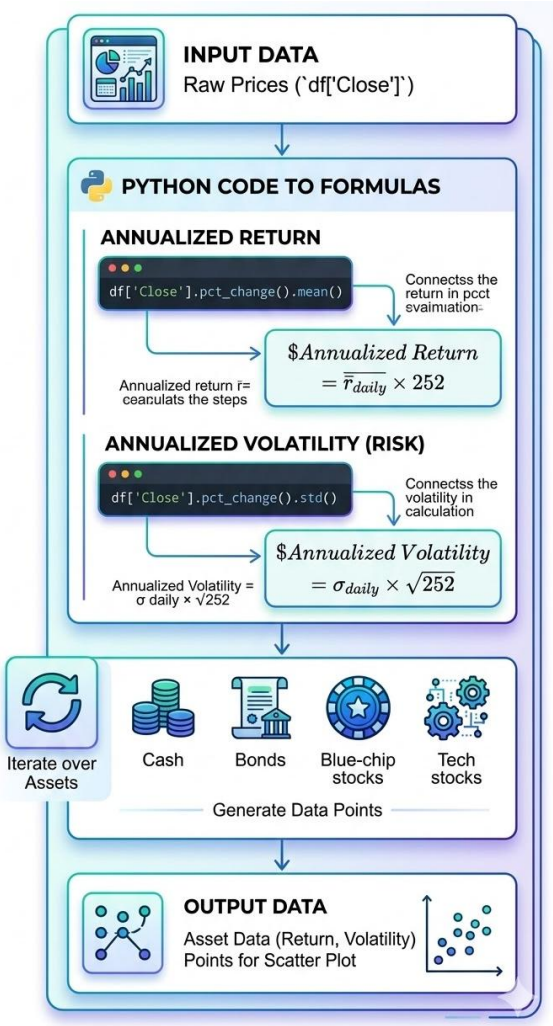
Plotting with mplfinance

- Standard matplotlib is difficult to use for specialized financial charts.
- We use mplfinance to automatically generate Candlesticks and Volume bars.
- **Volume:** The bar chart at the bottom shows how many shares changed hands that day. High volume validates a price movement; low volume suggests a lack of conviction.
- **Visualizing the Market:** You will see periods of tight, small candles (low volatility) and periods of massive, long red/green candles (market panics or booms).



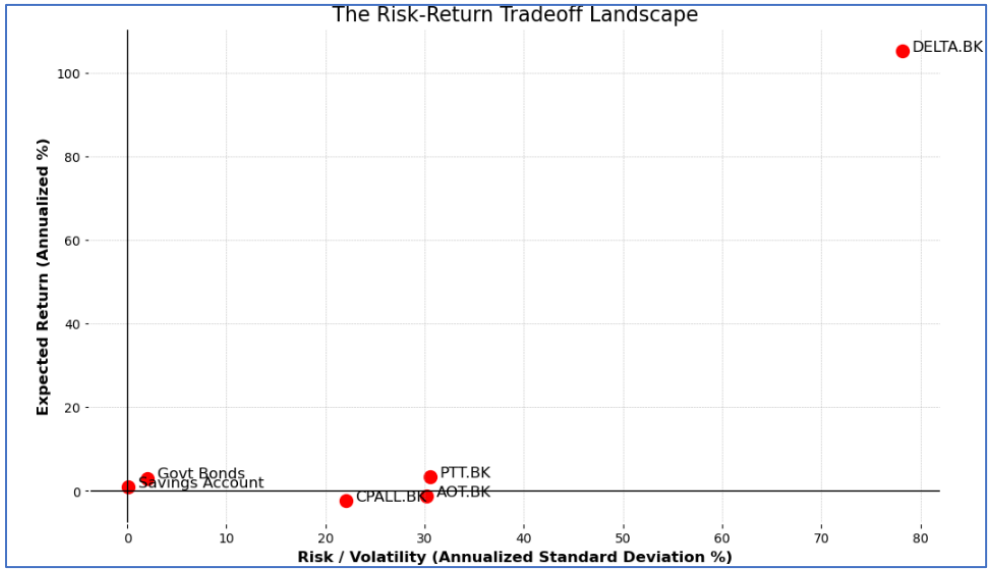
Calculating the Risk-Return Tradeoff in Code

- We convert raw prices into actionable metrics.
- **Daily Return:** df['Close'].pct_change()
- **Annualized Return:** daily_return.mean() * 252
- **Annualized Volatility (Risk):** daily_return.std() * np.sqrt(252)
- By running this loop across multiple assets (Cash, Bonds, Blue-chip stocks, Tech stocks), we generate the data points for our Risk-Return scatter plot.



Visualizing the Tradeoff Landscape

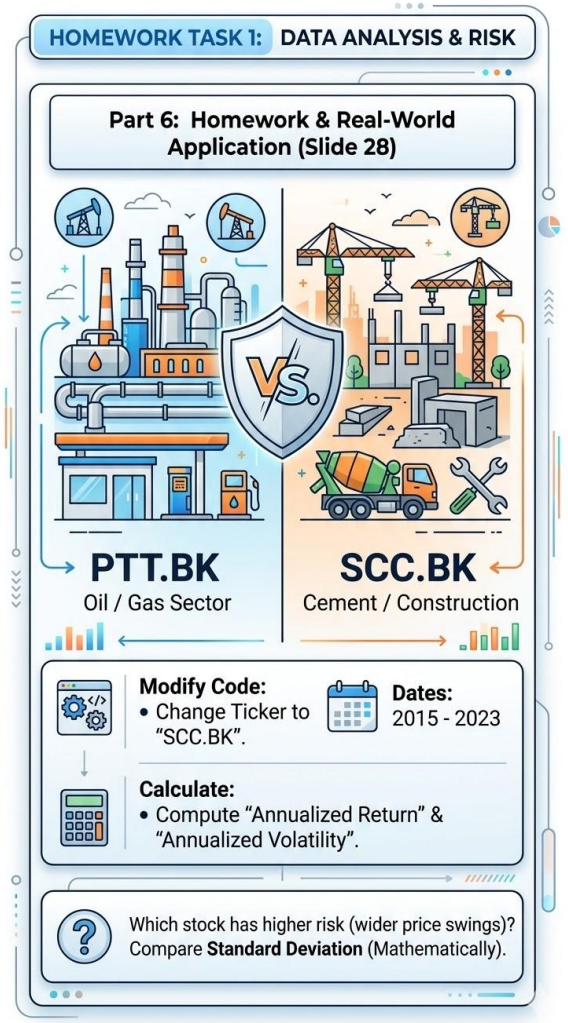
- The output of our code proves the economic theory.
- **The Scatter Plot:**
  - Bottom Left (Low Risk, Low Return): Savings Accounts, Government Bonds.
  - Middle (Moderate Risk/Return): PTT (Energy utility).
  - Top Right (High Risk, High Return): High-growth tech stocks.
- **The Lesson:** The code proves you cannot jump to the top of the Y-axis (Return) without dragging your point far out along the X-axis (Risk).



Part 6: Homework & Real-World Application

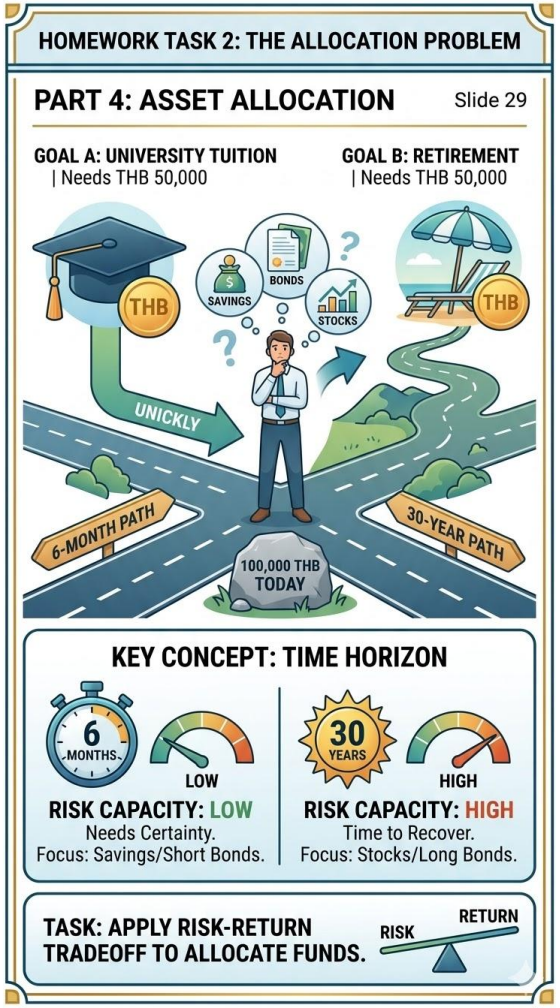
Homework Task 1: Data Analysis & Risk

- **Context:** In the tutorial, we analyzed PTT.BK. Now you must apply this to a different sector.
- **Task:** Modify the Python code to fetch data for Siam Cement Group (SCC.BK) from 2015 to 2023.
- **Calculation:** Compute its Annualized Return and Annualized Volatility.
- **Critical Question:** Compare SCC's standard deviation to PTT's. Mathematically, which stock exhibited higher risk (wider price swings) over this period?



Homework Task 2: The Allocation Problem

- **Scenario:** You have 100,000 THB today.
  - **Goal A:** 50,000 THB is needed in exactly 6 months to pay university tuition.
  - **Goal B:** 50,000 THB is saved for your retirement in 30 years.
- **Task:** Apply the Risk-Return Tradeoff to allocate these funds among Savings Accounts, Bonds, and Stocks.
- **Key Concept:** *Time Horizon.* Risk capacity is heavily dependent on when you need the money.



Conclusion & Course Wrap-Up

- **Capital Markets:** The engine of long-term economic funding (Debt vs. Equity).
- **Returns:** Investors seek Dividends (cash flow) and Capital Gains (speculation).
- **The Law of Finance:** High returns demand high volatility (Risk).
- **Data Science:** Python allows us to quantify, visualize, and prove these economic theories using real-time market data.



Let’s check out our Notebook

